3

Interdependence and the Gains from Trade

PowerPoint Slides prepared by:
Andreea CHIRITESCU
Eastern Illinois University

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Why We Interact with Others

- Why don't we "bother" to do so many things? Maybe not that we are not good at doing them, but it's not worthwhile…

Why doesn't Tiger Wood mow his lawn?

Why We Specialize & Trade

- Questions (Food for thought)
  - Why the majority of chefs are males, but then why they let their wives cook?
  - Why does your professor let a TA proctor the exam?

Why We Trade

- Do you want a job … on an assembly line?
  - What would be the opportunity cost of working on an assembly line?

<$1/hour

>$20/hour

A Parable for the Modern Economy

- Only two goods
  - Meat
  - Potatoes
- Only two people
  - Rancher
  - Farmer

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

A Parable for the Modern Economy

- If rancher produces only meat
  - And farmer produces only potatoes
  - Both gain from trade
- If both rancher and farmer produce both meat and potatoes
  - Both gain from specialization and trade
- *What if either one is better at producing both the goods?*
- Production possibilities frontier
  - Various mixes of output that an economy can produce

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Figure 1

The Production Possibilities Frontier (a)

(a) Production Opportunities

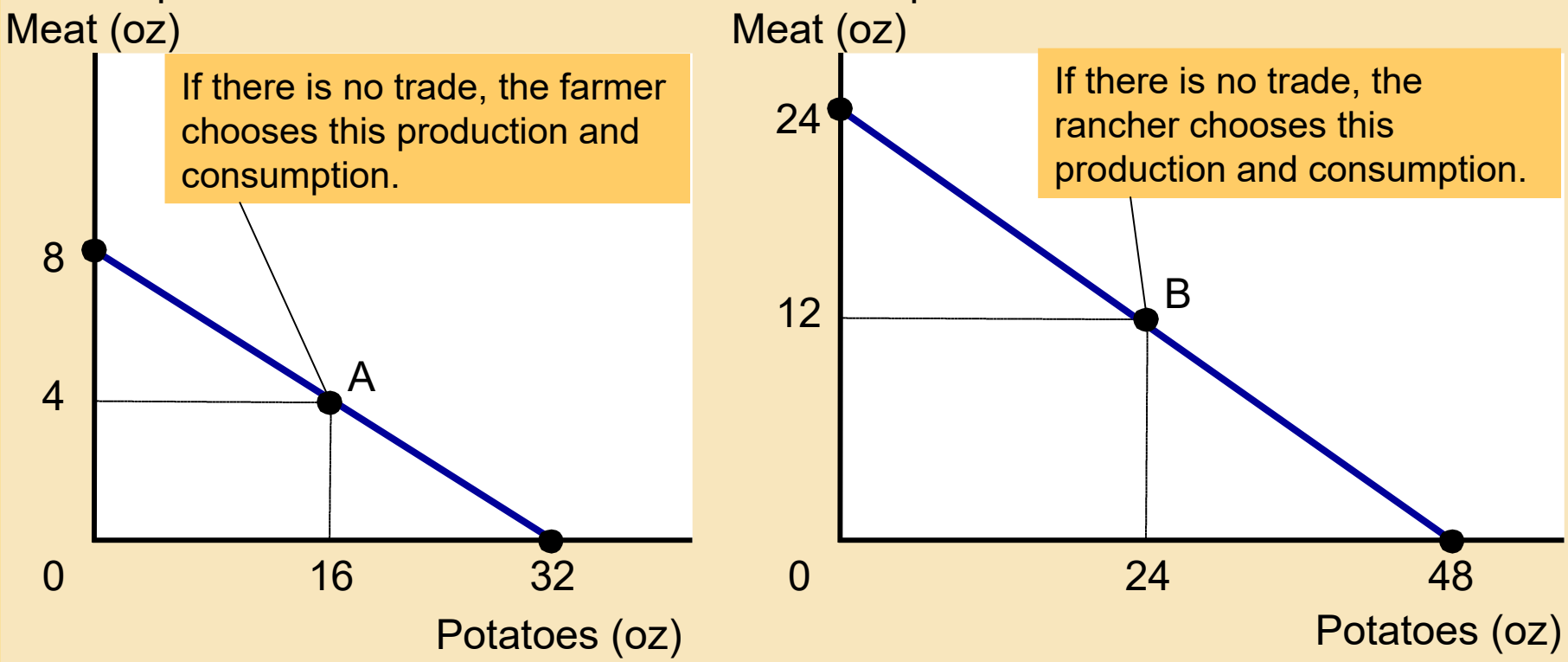
	Minutes Needed to Make 1 Ounce of:		Amount Produced in 8 Hours	
	Meat	Potatoes	Meat	Potatoes
Farmer	60 min/oz	15 min/oz	8 oz	32 oz
Rancher	20 min/oz	10 min/oz	24 oz	48 oz

Panel (a) shows the production opportunities available to the farmer and the rancher.

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Figure 1

The Production Possibilities Frontier (b, c)



Panel (b) shows the combinations of meat and potatoes that the farmer can produce. Panel (c) shows the combinations of meat and potatoes that the rancher can produce. Both production possibilities frontiers are derived assuming that the farmer and rancher each work 8 hours per day. If there is no trade, each person's production possibilities frontier is also his or her consumption possibilities frontier.

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

A Parable for the Modern Economy

- Specialization and trade
  - Farmer – specialize in growing potatoes
    - More time growing potatoes
    - Less time raising cattle
  - Rancher – specialize in raising cattle
    - More time raising cattle
    - Less time growing potatoes
  - Trade: 5 oz of meat for 15 oz of potatoes
  - Both gain from specialization and trade

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Figure 2

How Trade Expands the Set of Consumption Opportunities (c)

(c) The Gains from Trade: A Summary

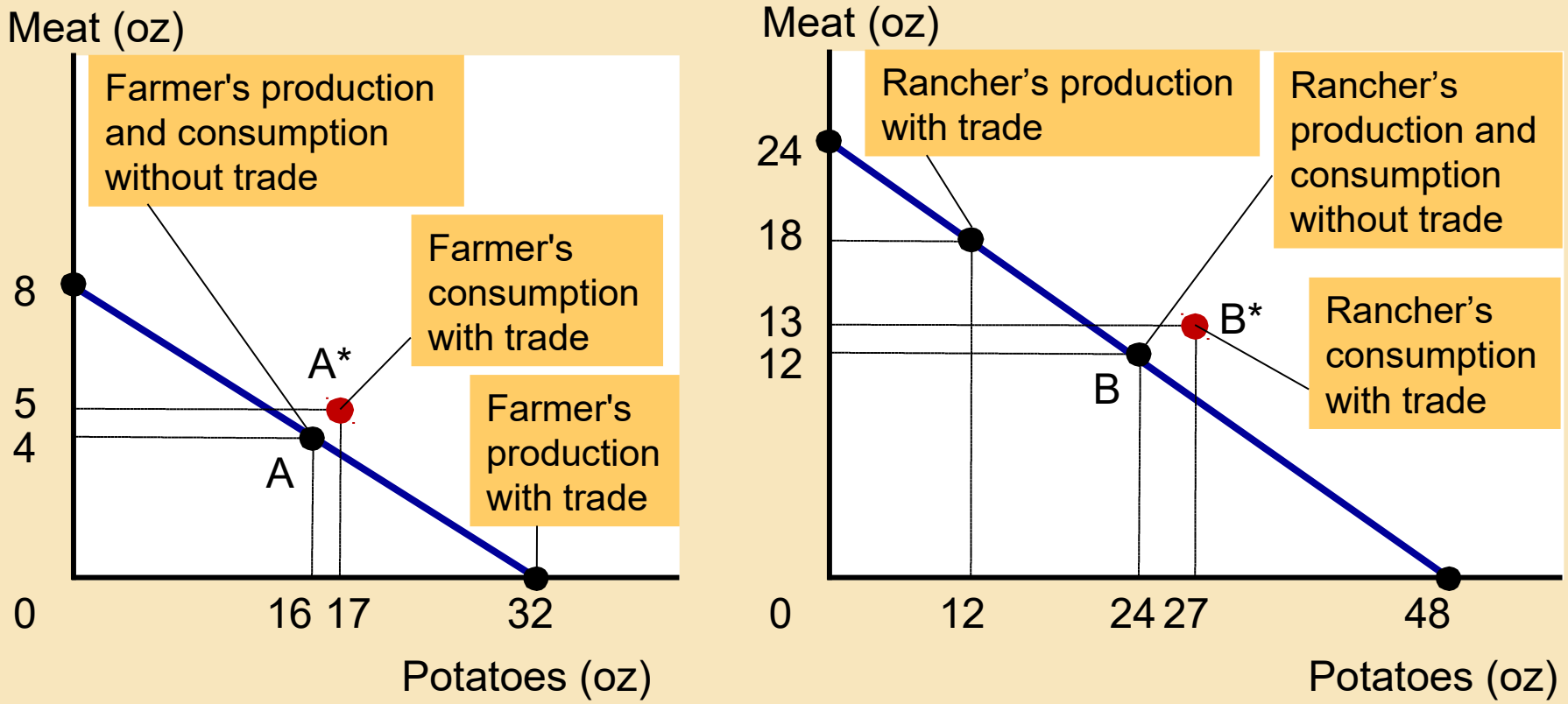
	Farmer		Rancher	
	Meat	Potatoes	Meat	Potatoes
Without Trade:				
Production and Consumption	4 oz	16 oz	12 oz	24 oz
With Trade:				
Production	0 oz	32 oz	18 oz	12 oz
Trade	Gets 5 oz	Gives 15 oz	Gives 5 oz	Gets 15 oz
Consumption	5 oz	17 oz	13 oz	27 oz
GAINS FROM TRADE:				
Increase in Consumption	+1 oz	+1 oz	+1 oz	+3 oz

The proposed trade between the farmer and the rancher offers each of them a combination of meat and potatoes that would be impossible in the absence of trade. In panel (a), the farmer gets to consume at point A* rather than point A. In panel (b), the rancher gets to consume at point B* rather than point B. Trade allows each to consume more meat and more potatoes.

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Figure 2

How Trade Expands the Set of Consumption Opportunities (a, b)



The proposed trade between the farmer and the rancher offers each of them a combination of meat and potatoes that would be impossible in the absence of trade. In panel (a), the farmer gets to consume at point A* rather than point A. In panel (b), the rancher gets to consume at point B* rather than point B. Trade allows each to consume more meat and more potatoes.

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Comparative Advantage

- Absolute advantage
  - Produce a good using fewer inputs than another producer
- Opportunity cost
  - Whatever must be given up to obtain some item
  - Measures the trade-off between the two goods that each producer faces

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Table 1

The Opportunity Cost of Meat and Potatoes

	Opportunity Cost of:	
	1 oz of Meat	1 oz of Potatoes
Farmer	4 oz potatoes	1/4 oz meat
Rancher	2 oz potatoes	1/2 oz meat

Who has a lower opportunity cost (comparative advantage) in producing meat?

Who has lower opportunity cost (comparative advantage) in producing potatoes?

Who should produce (specialize) only meat? Who should produce only potatoes?

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Comparative Advantage

- Comparative advantage
  - Produce a good at a lower opportunity cost than another producer
  - Reflects the relative opportunity cost
- Principle of comparative advantage
  - Each good - produced by the individual that has the smaller opportunity cost of producing that good

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Comparative Advantage

- One person
  - Can have absolute advantage in both goods
  - Cannot have comparative advantage in both goods
- For different opportunity costs
  - One person - comparative advantage in one good
  - The other person - comparative advantage in the other good

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Comparative Advantage

- Opportunity cost of one good
  - Inverse of the opportunity cost of the other
- Gains from specialization and trade
  - Based on comparative advantage
  - Total production in economy rises
    - Increase in the size of the economic pie
    - Everyone – better off

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Comparative Advantage

- Trade can benefit everyone in society
  - Allows people to specialize
- The price of trade
  - Must lie between the two opportunity costs
  - *3 in this case, 15/5 for the farmer, 1/3 for the rancher.*
- Principle of comparative advantage explains:
  - Interdependence
  - Gains from trade

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Applications of Comparative Advantage

- Should the U.S. trade with other countries?
  - U.S and Japan
    - Each produces food and cars
    - One American worker, one month
      - One car, or
      - Two tons of food
    - One Japanese worker, one month
      - One car
      - One ton of food

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Applications of Comparative Advantage

- Principle of comparative advantage
  - Each good – produced by the country with the smaller opportunity cost of producing that good
- Specialization and trade
  - All countries have more food and more cars

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.